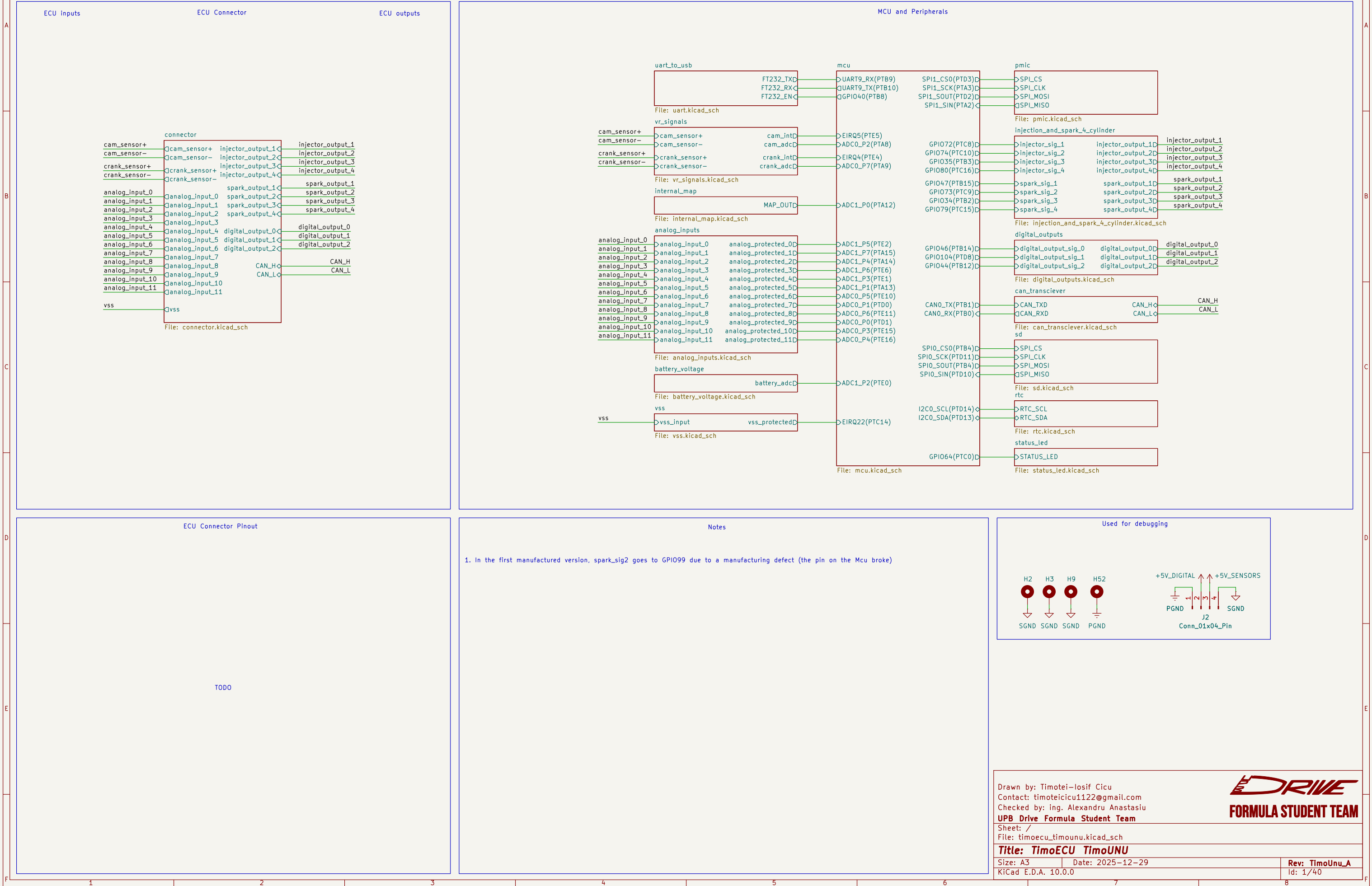
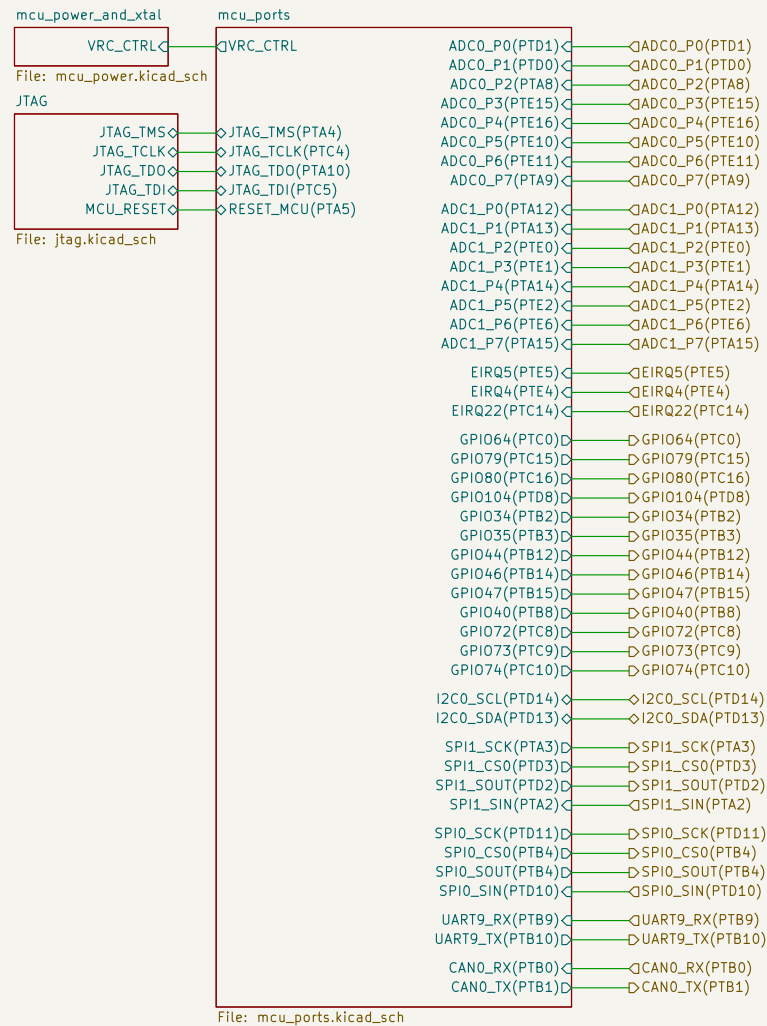


TimoEcu TimoUnu main schematic. (Block-diagram-like schematic)



Microcontroller with JTAG port



Drawn by: Timotei-Iosif Cicu
Contact: timoteicicu1122@gmail.com

Checked by:

UPB Drive Formula Student Team

Sheet: /mcu/
File: mcu.kicad_sch

Title: **TimoECU TimoUNU**

Size: A4

Date: 2025-12-29

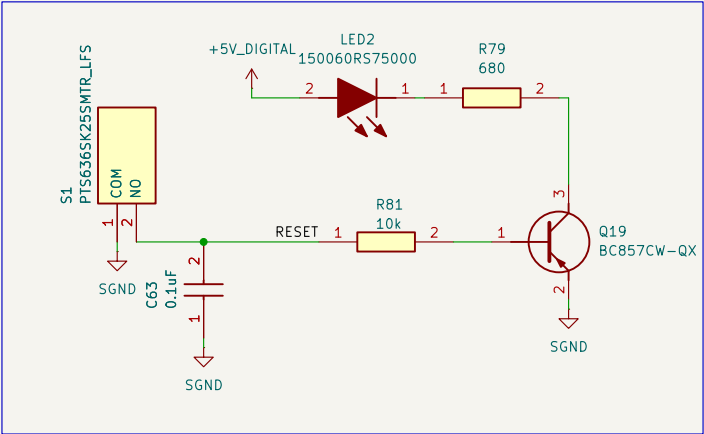
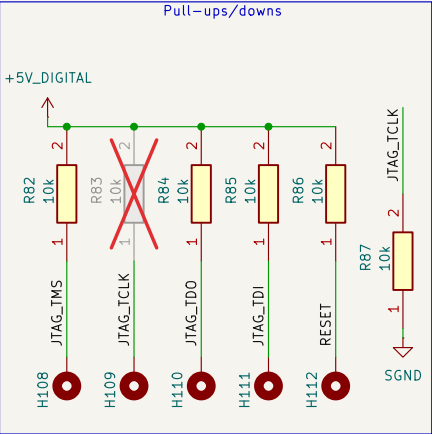
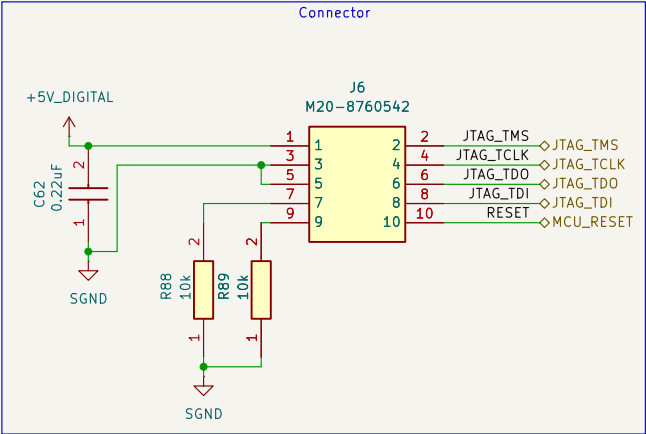
KiCad E.D.A. 10.0.0

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Rev: **TimoUnu_A**

Id: 2/40

Cortex Debug Connector



Reference from S32K312 EVB

Note: Reference missing in released schematic due to it being from a datasheet or another person's project

Drawn by: Timotei-Iosif Cicu
Contact: timoteicicu1122@gmail.com

UPB Drive Formula Student Team

Sheet: /mcu/JTAG/
File: jtag.kicad_sch

Title: **TimoECU TimoUNU**

Size: A4
KiCad E.D.A. 10.0.0

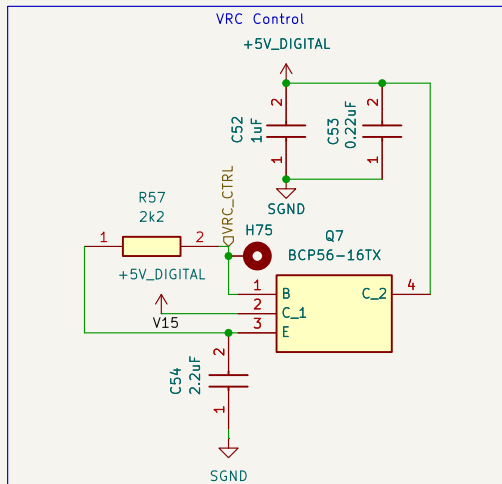
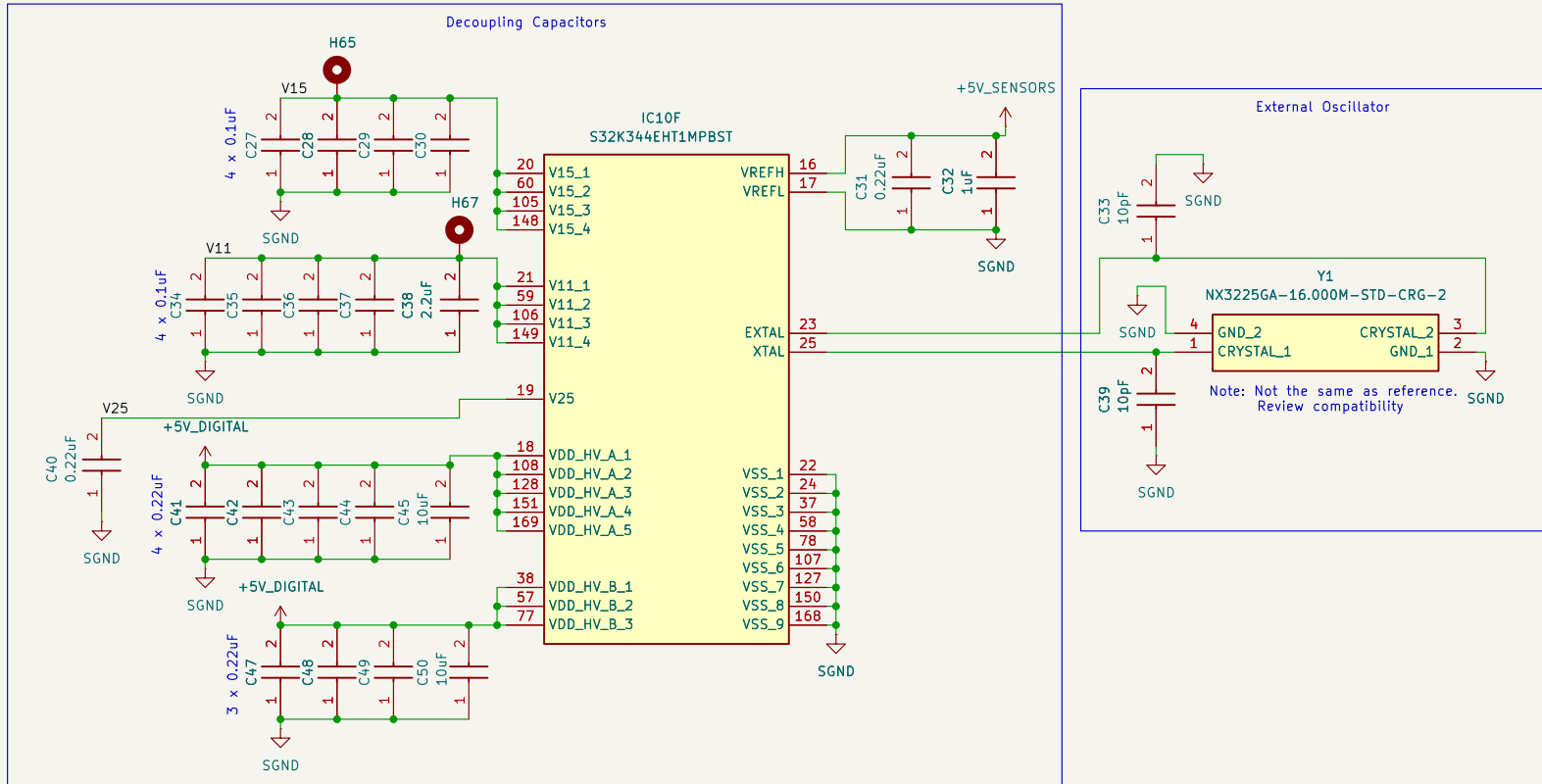
Date:

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Rev: **TimoUnu_A**
Id: 40/40

Reference: S32K344 MINI Eval Board schematic

MCU Power Pins



Drawn by: Timotei-Iosif Cicu
Contact: timoteicicu1122@gmail.com

UPB Drive Formula Student Team

Sheet: /mcu/mcu_power_and_xtal/
File: mcu_power.kicad_sch

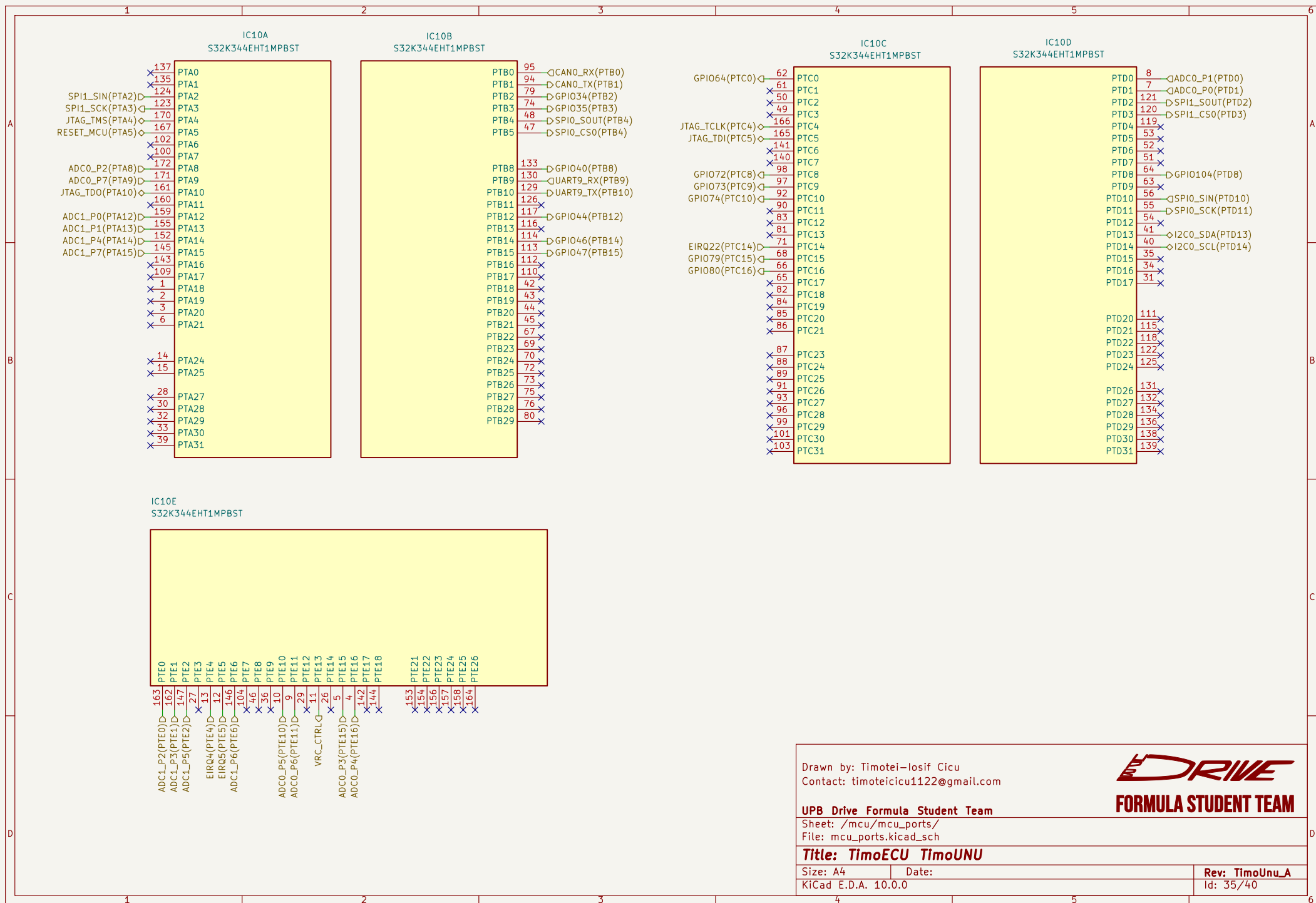
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Size: A4
KiCad E.D.A. 10.0.0

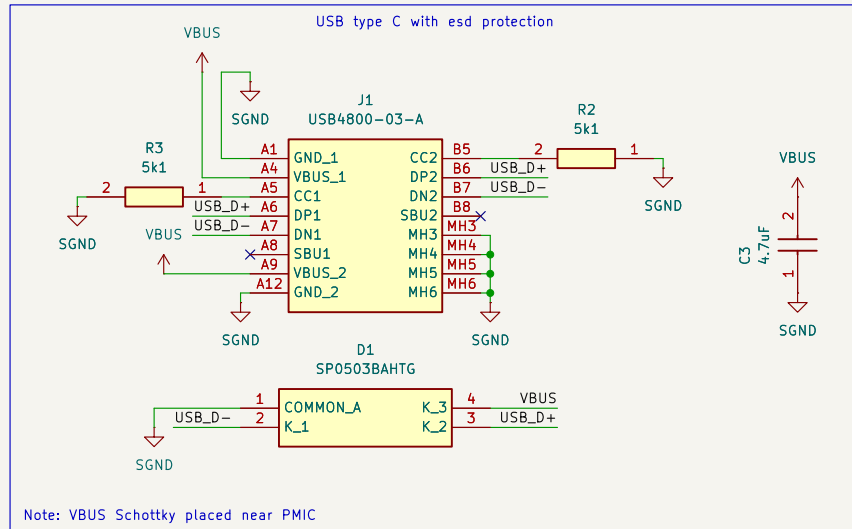
Date:

DRIVE
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Rev: TimoUnu_A
Id: 34/40

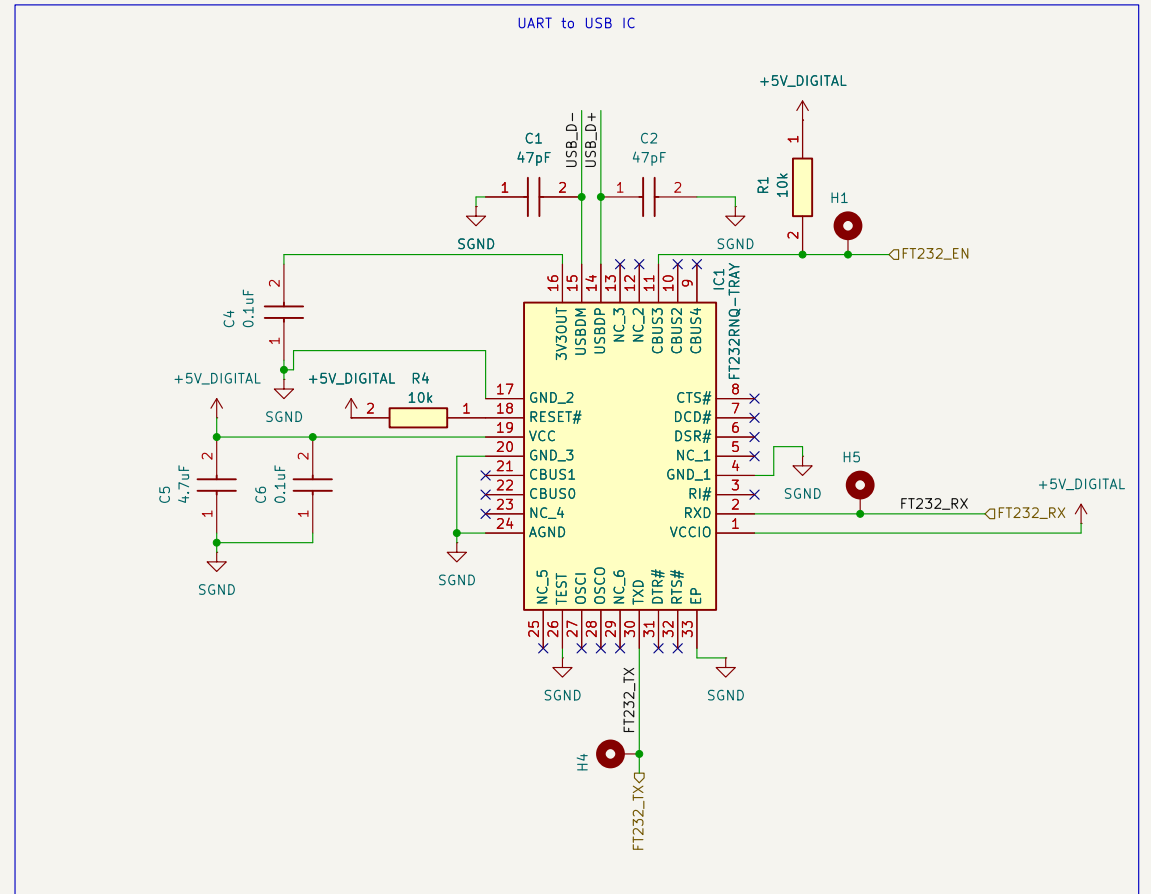


UART to USB



Reference from FT232RN datasheet

Note: Reference missing in released schematic due to it being from a datasheet or another person's project



Drawn by: Timotei-Iosif Cicu
Contact: timoteicicu1122@gmail.com
Checked by:
UPB Drive Formula Student Team
Sheet: /uart_to_usb/
File: uart.kicad_sch

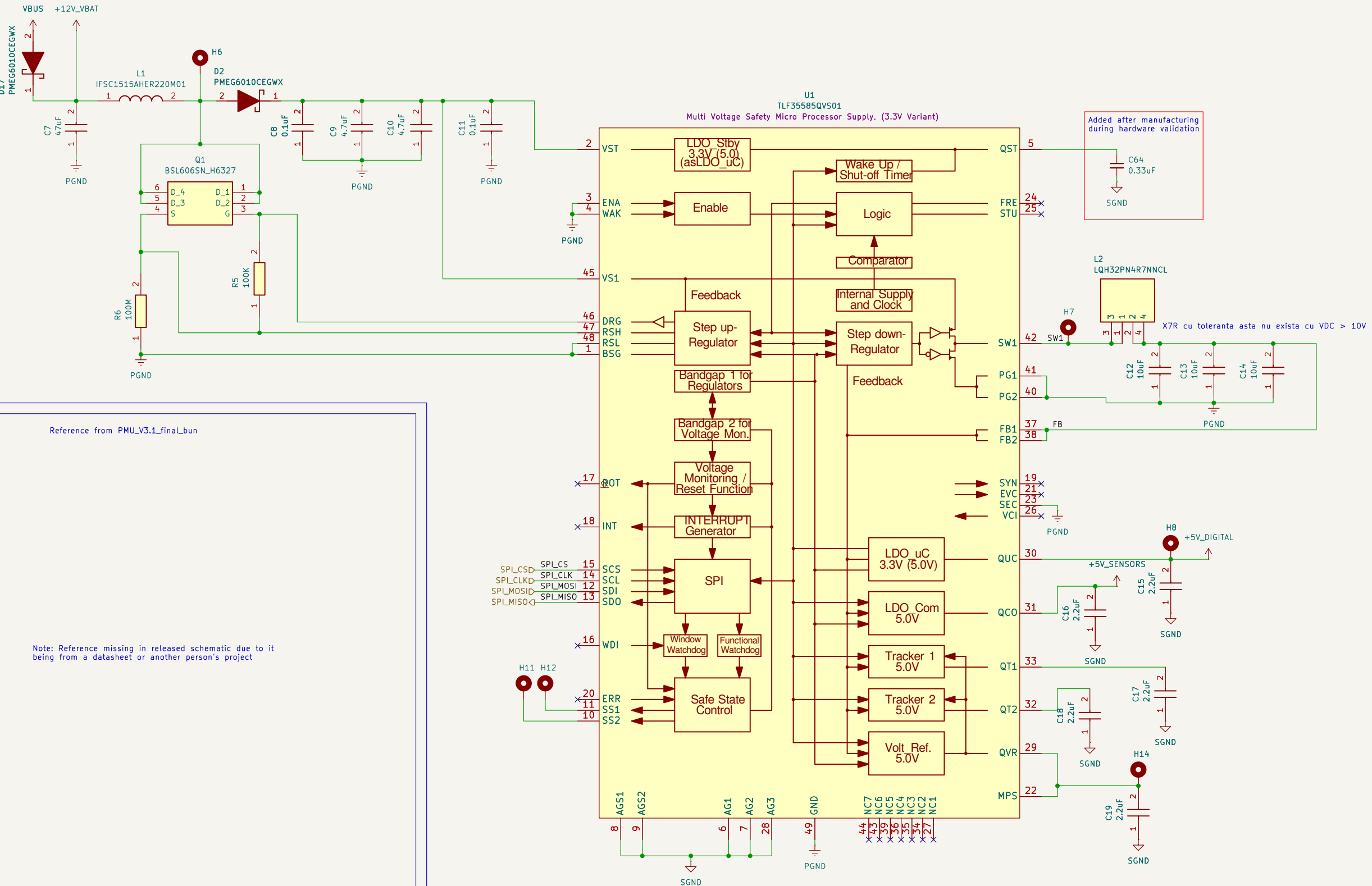
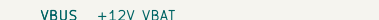
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Size: A4 Date: 2025-12-29
KiCad E.D.A. 10.0.0

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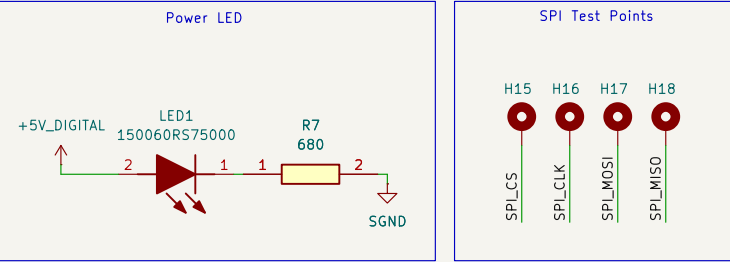
Rev: **TimoUnu_A**
Id: 3/40

Power Management IC



Reference from PMU_V3.1_final_bun

Note: Reference missing in released schematic due to it being from a datasheet or another person's project



Drawn by: Timotei-Iosif Cicu
Contact: timoteicicu1122@gmail.com
Checked by:
UPB Drive Formula Student Team
Sheet: /pmic/
File: pmic.kicad_sch



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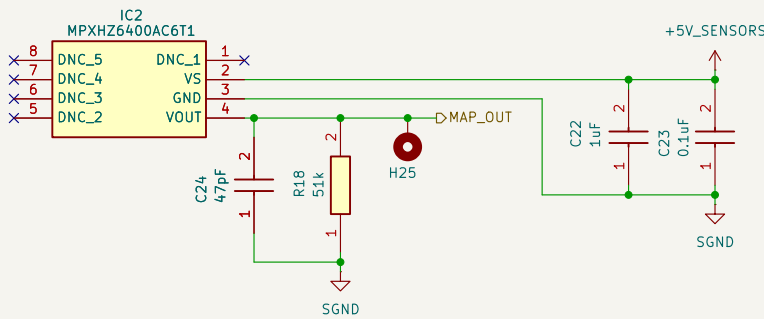
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Size: A3	Date: 2025-12-29
KiCad E.D.A. 10.0.0	

Rev: TimoUnu_A
d: 4/40

On-board MAP sensor

KiCad Schematic



Reference from datasheet

Note: Reference missing in released schematic due to it being from a datasheet or another person's project

Drawn by: Timotei-Iosif Cicu
Contact: timoteicicu1122@gmail.com
Checked by:
UPB Drive Formula Student Team
Sheet: /internal_map/
File: internal_map.kicad_sch

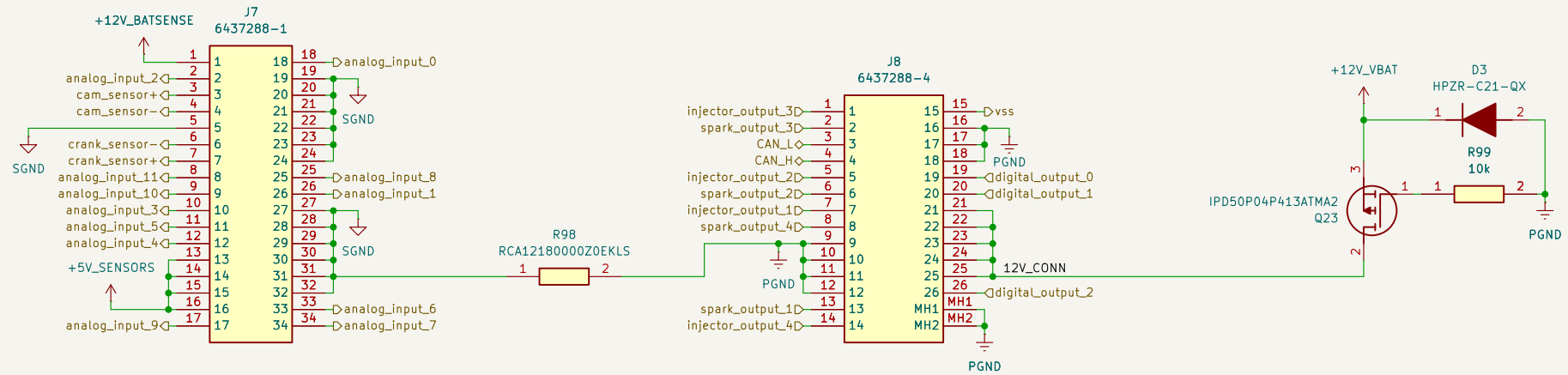


Title: TimoECU TimoUNU

Size: A4
Date: 2025-12-29
KiCad E.D.A. 10.0.0

Rev: TimoUnu_A
Id: 8/40

ECU Connector



+ 4 for VR signals
+ 12 for Analog inputs
+ 1 for VSS
+ 8 for spark and injection
+ 4 for digital outputs
+ 2 for CANH and CANL
+ 4 or more for future developments

Drawn by: Timotei-Iosif Cicu
Contact: timoteicicu1122@gmail.com
Checked by:

UPB Drive Formula Student Team

Sheet: /connector/

File: connector.kicad_sch

Title: **TimoECU TimoUNU**

Size: A4 Date: 2025-12-29

KiCad E.D.A. 10.0.0

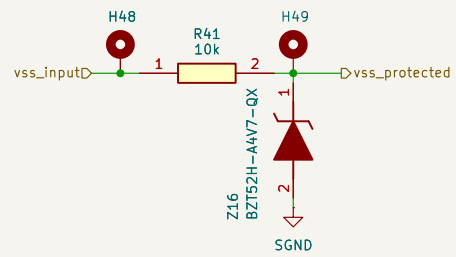
DRIVE
FORMULA STUDENT TEAM

Rev: **TimoUnu_A**

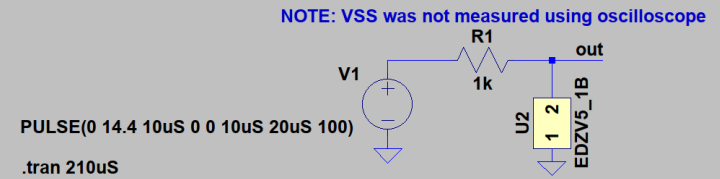
Id: 20/40

VSS input with overvoltage protection

KiCad Schematic



LTSpice Schematic



Drawn by: Timotei-Iosif Cicu
Contact: timoteicicu1122@gmail.com
Checked by:
UPB Drive Formula Student Team
Sheet: /vss/
File: vss.kicad_sch

Title: TimoECU TimoUNU

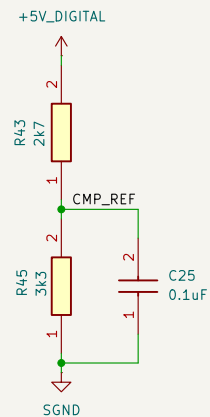
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KiCad E.D.A. 10.0.0

DRIVE
FORMULA STUDENT TEAM

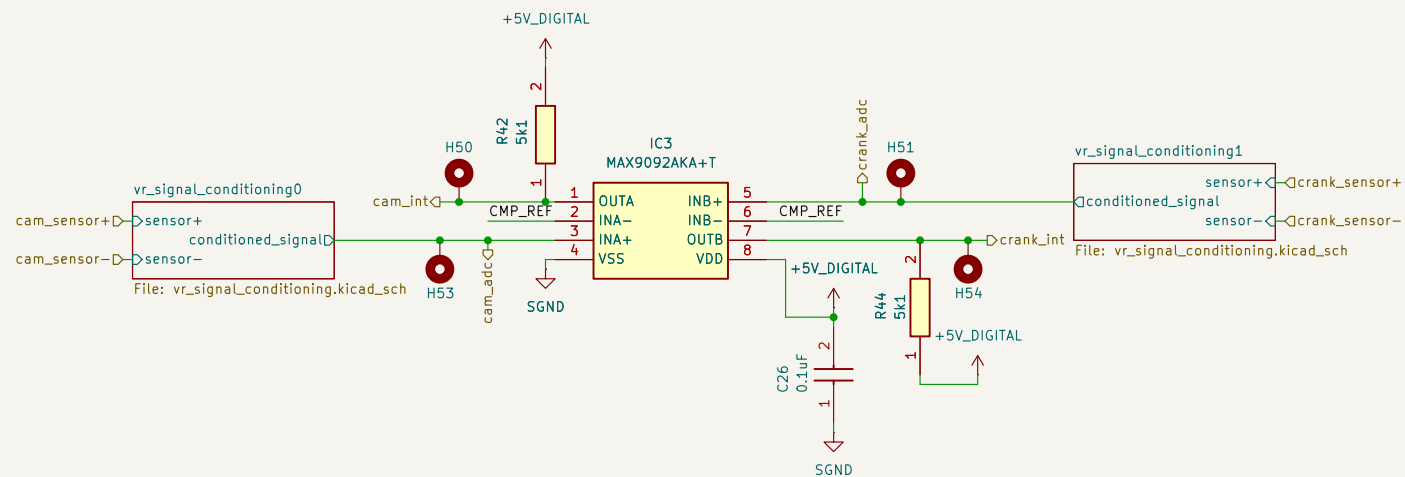
Rev: TimoUnu_A
Id: 21/40

Crank and Cam circuit

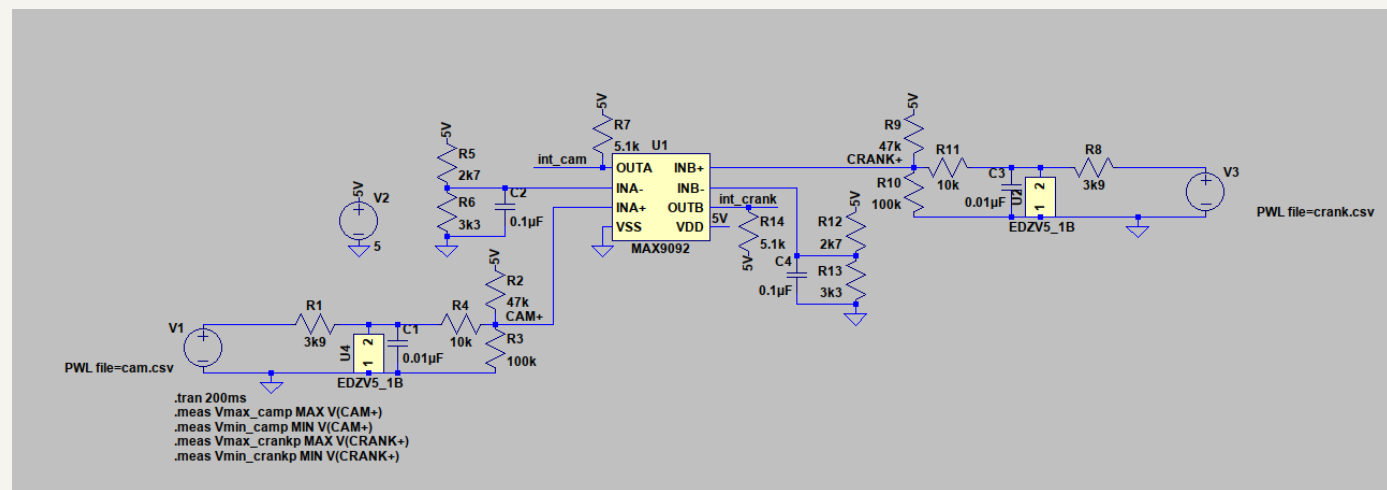
Comparator Reference



Comparator circuit



LTSpice Simulation Schematic



Drawn by: Timotei-Iosif Cicu
Contact: timoteicicu1122@gmail.com
Checked by:

UPB Drive Formula Student Team

Sheet: /vr_signals/

File: vr_signals.kicad_sch

Title: TimoECU TimoUNU

Size: A4

Date: 2025-12-29

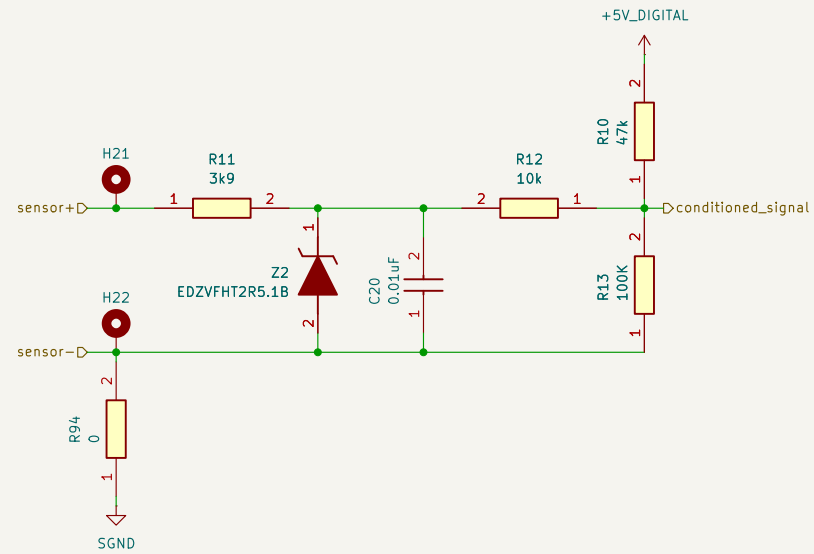
KiCad E.D.A. 10.0.0

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Rev: TimoUnu_A

Id: 22/40

VR Signal Conditioning



Drawn by: Timotei-Iosif Cicu
 Contact: timoteicicu1122@gmail.com
 Checked by:

UPB Drive Formula Student Team

Sheet: /vr_signals/vr_signal_conditioning0/
 File: vr_signal_conditioning.kicad_sch

Title: TimoECU TimoUNU

Size: A4

Date: 2025-12-29

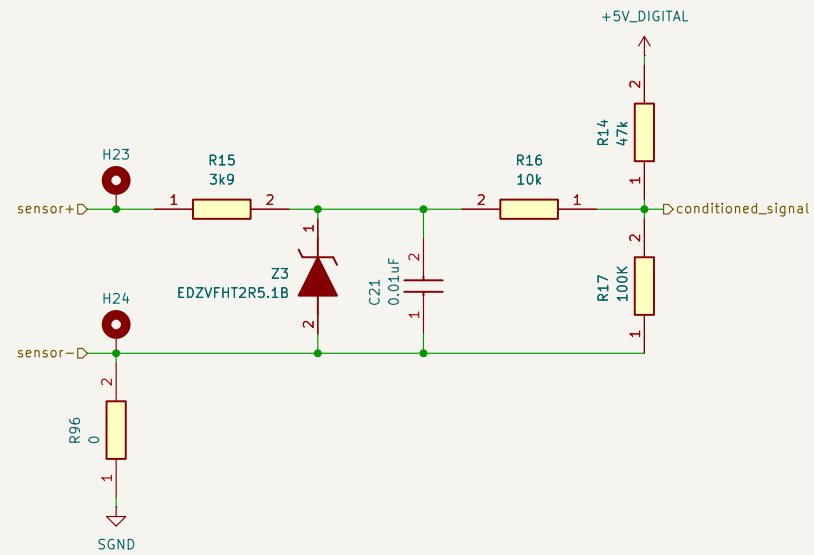
KiCad E.D.A. 10.0.0



Rev: TimoUnu_A

Id: 6/40

VR Signal Conditioning



Drawn by: Timotei-Iosif Cicu
 Contact: timoteicicu1122@gmail.com
 Checked by:

UPB Drive Formula Student Team

Sheet: /vr_signals/vr_signal_conditioning1/
 File: vr_signal_conditioning.kicad_sch

Title: TimoECU TimoUNU

Size: A4 Date: 2025-12-29
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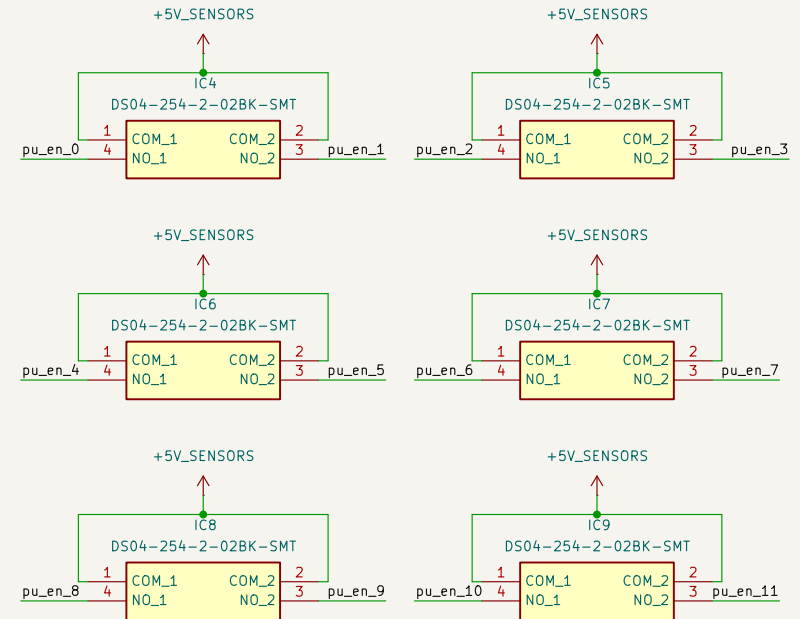
Rev: TimoUnu_A
 Id: 7/40

Analog Inputs

Analog inputs subschematics



Configurable pull-ups



Drawn by: Timotei-Iosif Cicu
 Contact: timoteicicu1122@gmail.com
 Checked by:
UPB Drive Formula Student Team
 Sheet: /analog_inputs/
 File: analog_inputs.kicad_sch

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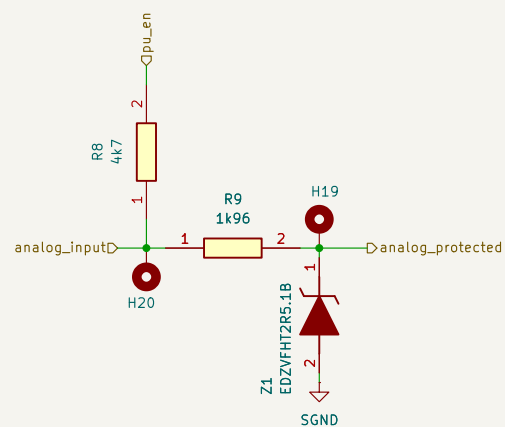
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Rev: TimoUnu_A
 Id: 23/40

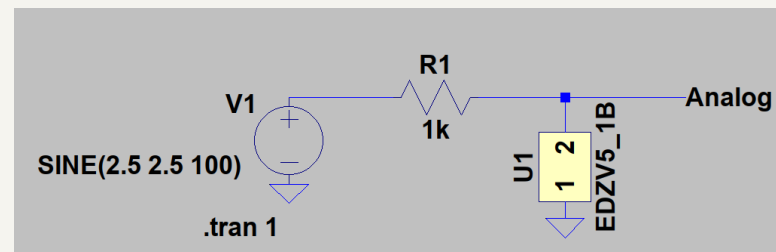
Single analog input block with protections.

KiCad Schematic



Note: $P_d = 55\text{mW}$ on Zener at 21V

LTSpice Schematic



Drawn by: Timotei-Iosif Cicu
Contact: timoteicicu1122@gmail.com
Checked by:

UPB Drive Formula Student Team

Sheet: /analog_inputs/analog_input0/

File: analog_input.kicad_sch

Title: TimoECU TimoUNU

Size: A4 Date: 2025-12-29

KiCad E.D.A. 10.0.0

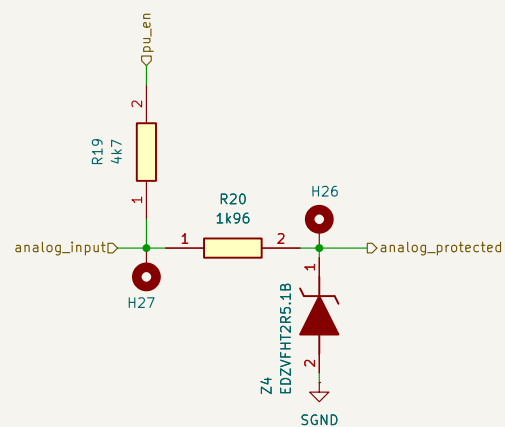
DRIVE
FORMULA STUDENT TEAM

Rev: TimoUnu_A

Id: 5/40

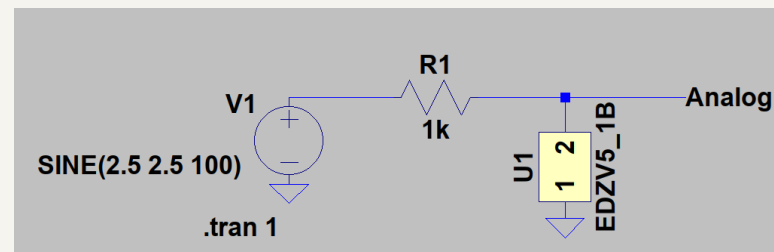
Single analog input block with protections.

KiCad Schematic



Note: Pd = 55mW on Zener at 21V

LTSpice Schematic



Drawn by: Timotei-Iosif Cicu
Contact: timoteicicu1122@gmail.com
Checked by:

UPB Drive Formula Student Team

Sheet: /analog_inputs/analog_input1/

File: analog_input.kicad_sch

Title: TimoECU TimoUNU

Size: A4

Date: 2025-12-29

KiCad E.D.A. 10.0.0

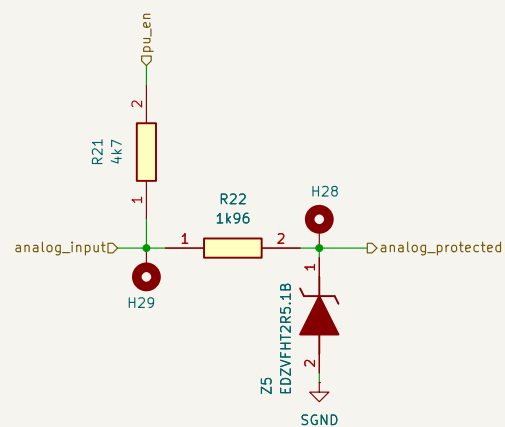
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FORMULA STUDENT TEAM

Rev: TimoUnu_A

Id: 9/40

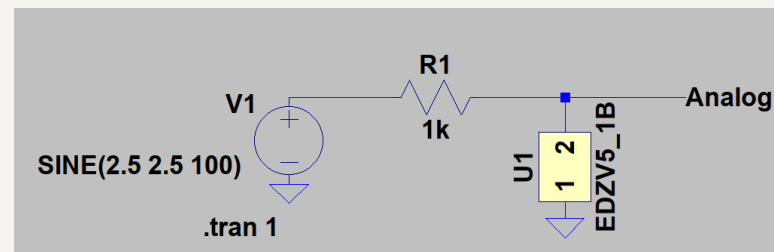
Single analog input block with protections.

KiCad Schematic



Note: Pd = 55mW on Zener at 21V

LTSpice Schematic



Drawn by: Timotei-Iosif Cicu
Contact: timoteicicu1122@gmail.com
Checked by:

UPB Drive Formula Student Team

Sheet: /analog_inputs/analog_input2/

File: analog_input.kicad_sch

Title: TimoECU TimoUNU

Size: A4

Date: 2025-12-29

KiCad E.D.A. 10.0.0

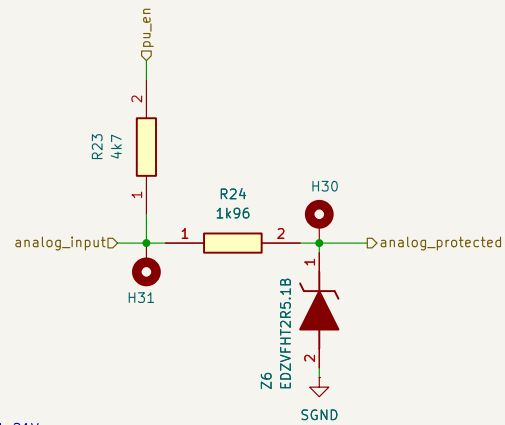
DRIVE
FORMULA STUDENT TEAM

Rev: TimoUnu_A

Id: 10/40

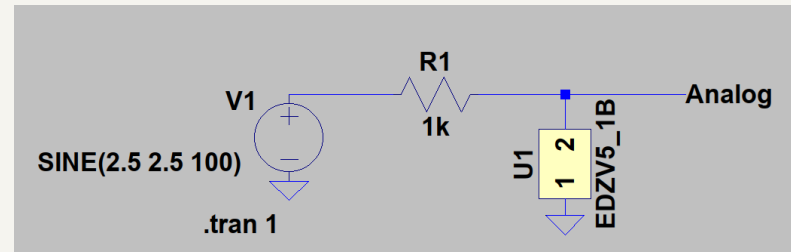
Single analog input block with protections.

KiCad Schematic



Note: $P_d = 55\text{mW}$ on Zener at 21V

LTSpice Schematic



Drawn by: Timotei-Iosif Cicu
Contact: timoteicicu1122@gmail.com
Checked by:

UPB Drive Formula Student Team

Sheet: /analog_inputs/analog_input3/
File: analog_input.kicad_sch

Title: TimoECU TimoUNU

Size: A4 Date: 2025-12-29

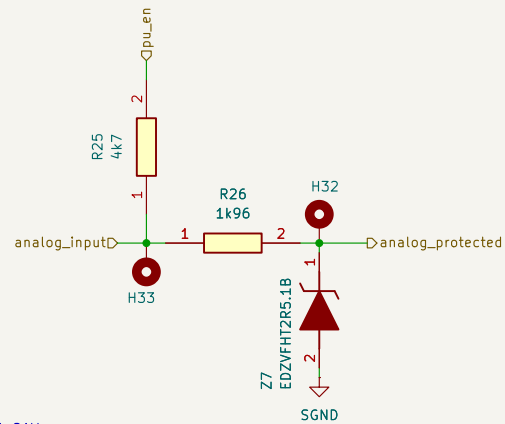
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Rev: TimoUnu_A
Id: 11/40

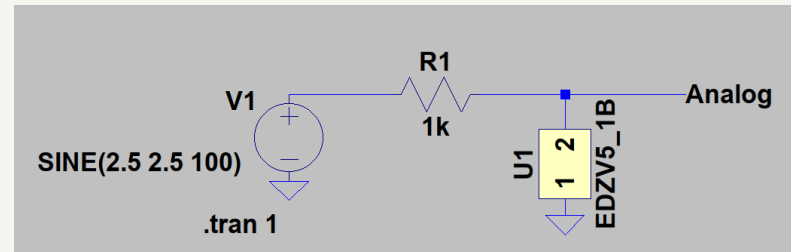
Single analog input block with protections.

KiCad Schematic



Note: Pd = 55mW on Zener at 21V

LTSpice Schematic



Drawn by: Timotei-Iosif Cicu
Contact: timoteicicu1122@gmail.com
Checked by:

UPB Drive Formula Student Team

Sheet: /analog_inputs/analog_input4/
File: analog_input.kicad_sch

Title: TimoECU TimoUNU

Size: A4 Date: 2025-12-29

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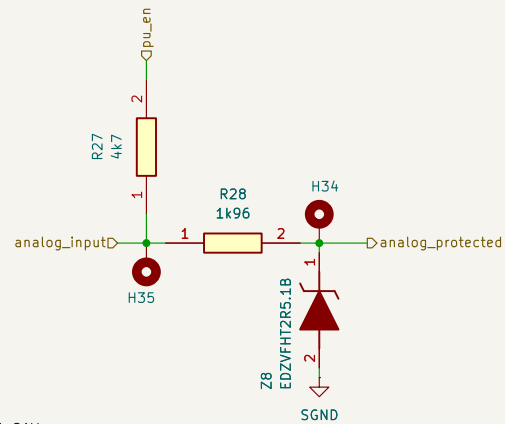
DRIVE
FORMULA STUDENT TEAM

Rev: TimoUnu_A

Id: 12/40

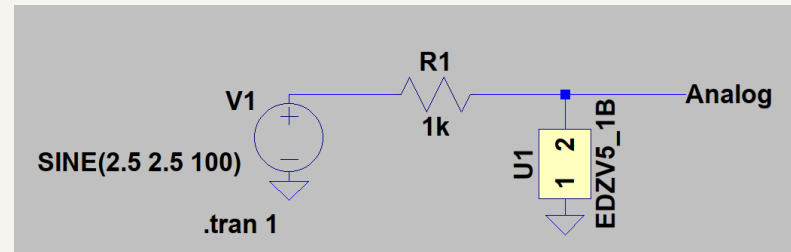
Single analog input block with protections.

KiCad Schematic



Note: Pd = 55mW on Zener at 21V

LTSpice Schematic



Drawn by: Timotei-Iosif Cicu
Contact: timoteicicu1122@gmail.com
Checked by:

UPB Drive Formula Student Team

Sheet: /analog_inputs/analog_input5/
File: analog_input.kicad_sch

Title: TimoECU TimoUNU

Size: A4 Date: 2025-12-29

KiCad E.D.A. 10.0.0

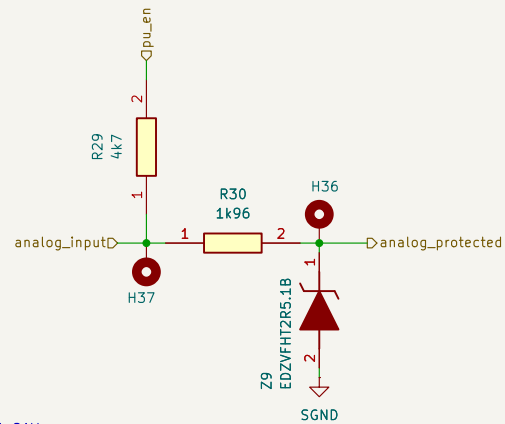
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FORMULA STUDENT TEAM

Rev: TimoUnu_A

Id: 13/40

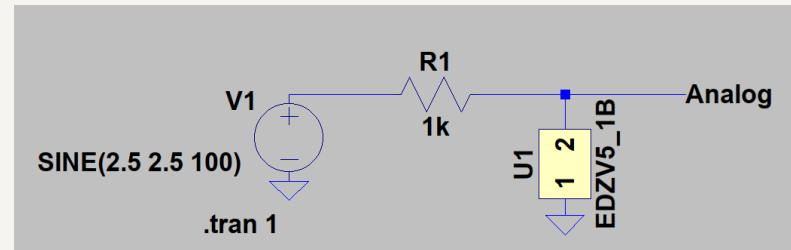
Single analog input block with protections.

KiCad Schematic



Note: $P_d = 55\text{mW}$ on Zener at 21V

LTSpice Schematic



Drawn by: Timotei-Iosif Cicu
Contact: timoteicicu1122@gmail.com
Checked by:

UPB Drive Formula Student Team

Sheet: /analog_inputs/analog_input6/
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Title: TimoECU TimoUNU

Size: A4 Date: 2025-12-29

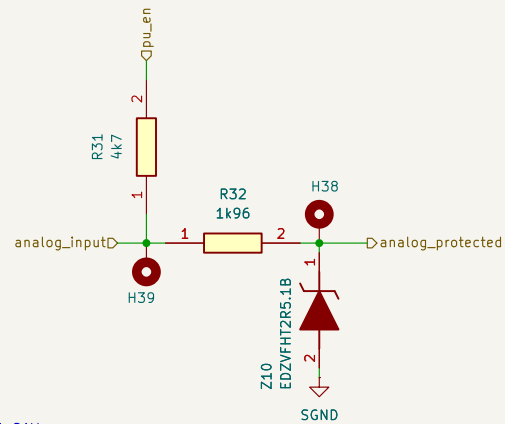
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Rev: TimoUnu_A
Id: 14/40

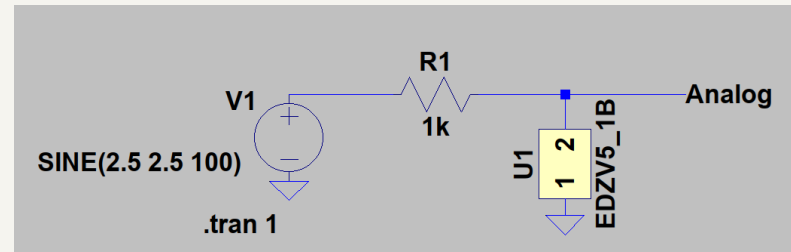
Single analog input block with protections.

KiCad Schematic



Note: Pd = 55mW on Zener at 21V

LTSpice Schematic



Drawn by: Timotei-Iosif Cicu
Contact: timoteicicu1122@gmail.com
Checked by:

UPB Drive Formula Student Team

Sheet: /analog_inputs/analog_input7/
File: analog_input.kicad_sch

Title: TimoECU TimoUNU

Size: A4 Date: 2025-12-29

KiCad E.D.A. 10.0.0

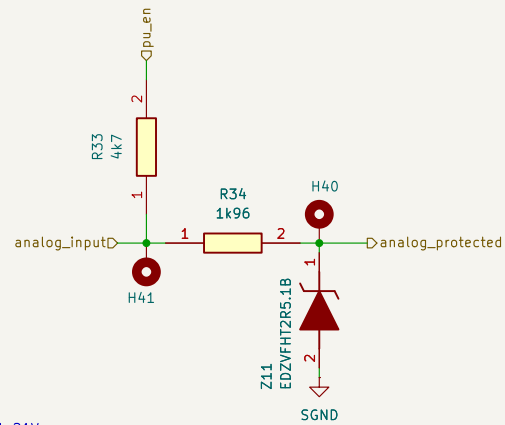
DRIVE
FORMULA STUDENT TEAM

Rev: TimoUnu_A

Id: 15/40

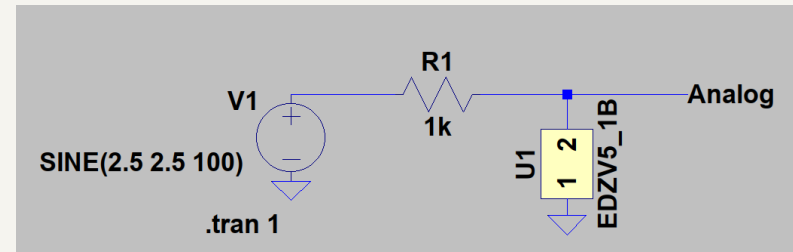
Single analog input block with protections.

KiCad Schematic



Note: Pd = 55mW on Zener at 21V

LTSpice Schematic



Drawn by: Timotei-Iosif Cicu
Contact: timoteicicu1122@gmail.com
Checked by:

UPB Drive Formula Student Team

Sheet: /analog_inputs/analog_input8/
File: analog_input.kicad_sch

Title: TimoECU TimoUNU

Size: A4 Date: 2025-12-29

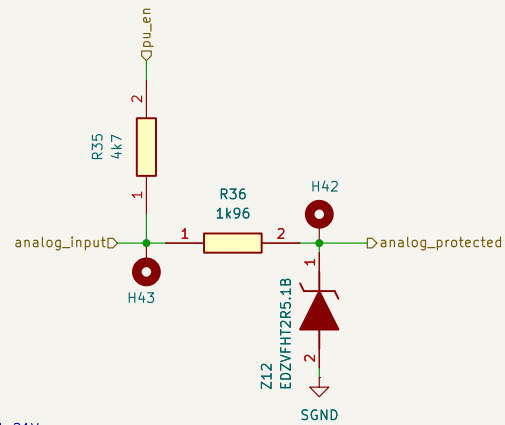
KiCad E.D.A. 10.0.0

DRIVE
FORMULA STUDENT TEAM

Rev: TimoUnu_A
Id: 16/40

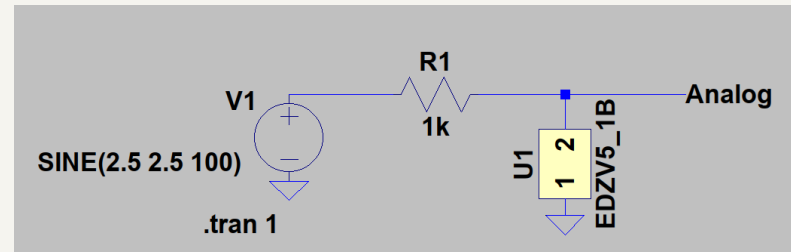
Single analog input block with protections.

KiCad Schematic



Note: Pd = 55mW on Zener at 21V

LTSpice Schematic



Drawn by: Timotei-Iosif Cicu
Contact: timoteicicu1122@gmail.com
Checked by:

UPB Drive Formula Student Team

Sheet: /analog_inputs/analog_input9/
File: analog_input.kicad_sch

Title: TimoECU TimoUNU

Size: A4 Date: 2025-12-29

KiCad E.D.A. 10.0.0

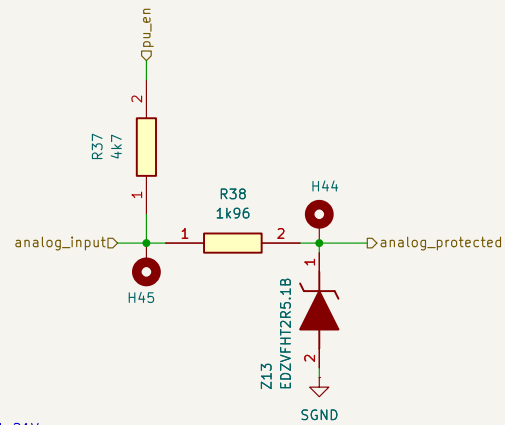
DRIVE
FORMULA STUDENT TEAM

Rev: TimoUnu_A

Id: 17/40

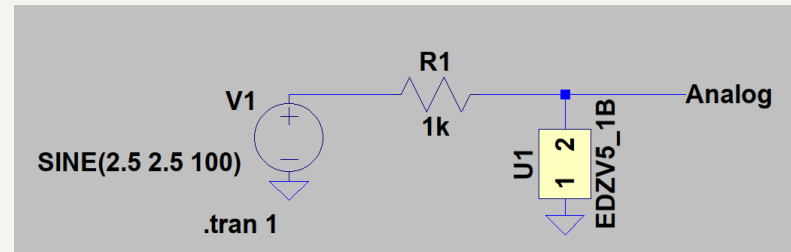
Single analog input block with protections.

KiCad Schematic



Note: Pd = 55mW on Zener at 21V

LTSpice Schematic



Drawn by: Timotei-Iosif Cicu
Contact: timoteicicu1122@gmail.com
Checked by:

UPB Drive Formula Student Team

Sheet: /analog_inputs/analog_input10/
File: analog_input.kicad_sch

Title: TimoECU TimoUNU

Size: A4 Date: 2025-12-29

KiCad E.D.A. 10.0.0

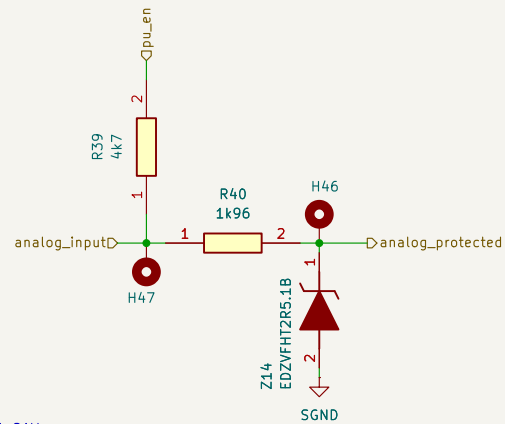
DRIVE
FORMULA STUDENT TEAM

Rev: TimoUnu_A

Id: 18/40

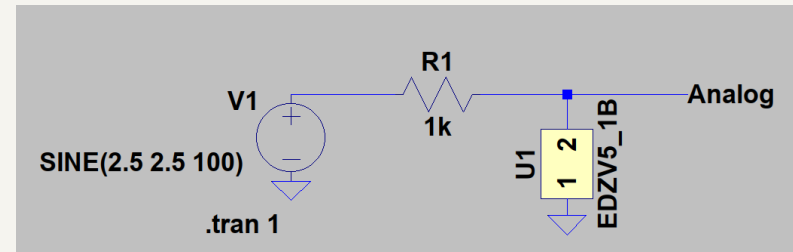
Single analog input block with protections.

KiCad Schematic



Note: Pd = 55mW on Zener at 21V

LTSpice Schematic



Drawn by: Timotei-Iosif Cicu
Contact: timoteicicu1122@gmail.com
Checked by:

UPB Drive Formula Student Team

Sheet: /analog_inputs/analog_input11/
File: analog_input.kicad_sch

Title: TimoECU TimoUNU

Size: A4 Date: 2025-12-29

KiCad E.D.A. 10.0.0

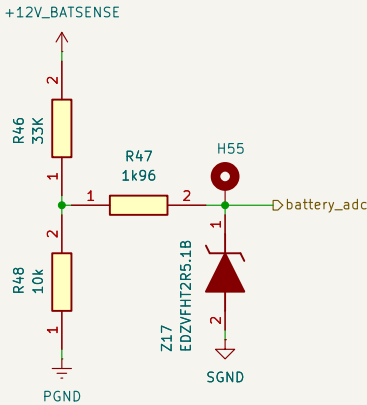
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Rev: TimoUnu_A

Id: 19/40

Battery Voltage detection trough voltage divider

Any battery voltage between 0V and 21.5V gets translated using the voltage divider into the 0V to 5V interval

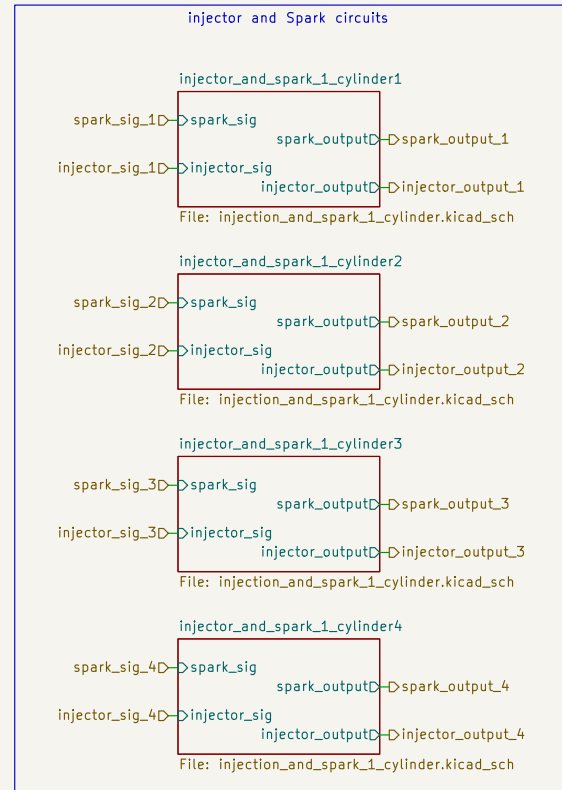


Drawn by: Timotei-Iosif Cicu
Contact: timoteicicu1122@gmail.com
Checked by:
UPB Drive Formula Student Team
Sheet: /battery_voltage/
File: battery_voltage.kicad_sch



Title: TimoECU TimoUNU		
Size: A4	Date: 2025-12-29	Rev: TimoUnu_A
KiCad E.D.A. 10.0.0	Id: 24/40	

injector and Spark for 4 cylinders



Drawn by: Timotei-Iosif Cicu
Contact: timoteicicu1122@gmail.com
Checked by:

UPB Drive Formula Student Team

Sheet: /injection_and_spark_4_cylinder/
File: injection_and_spark_4_cylinder.kicad_sch

Title: TimoECU TimoUNU

Size: A4

Date: 2025-12-29

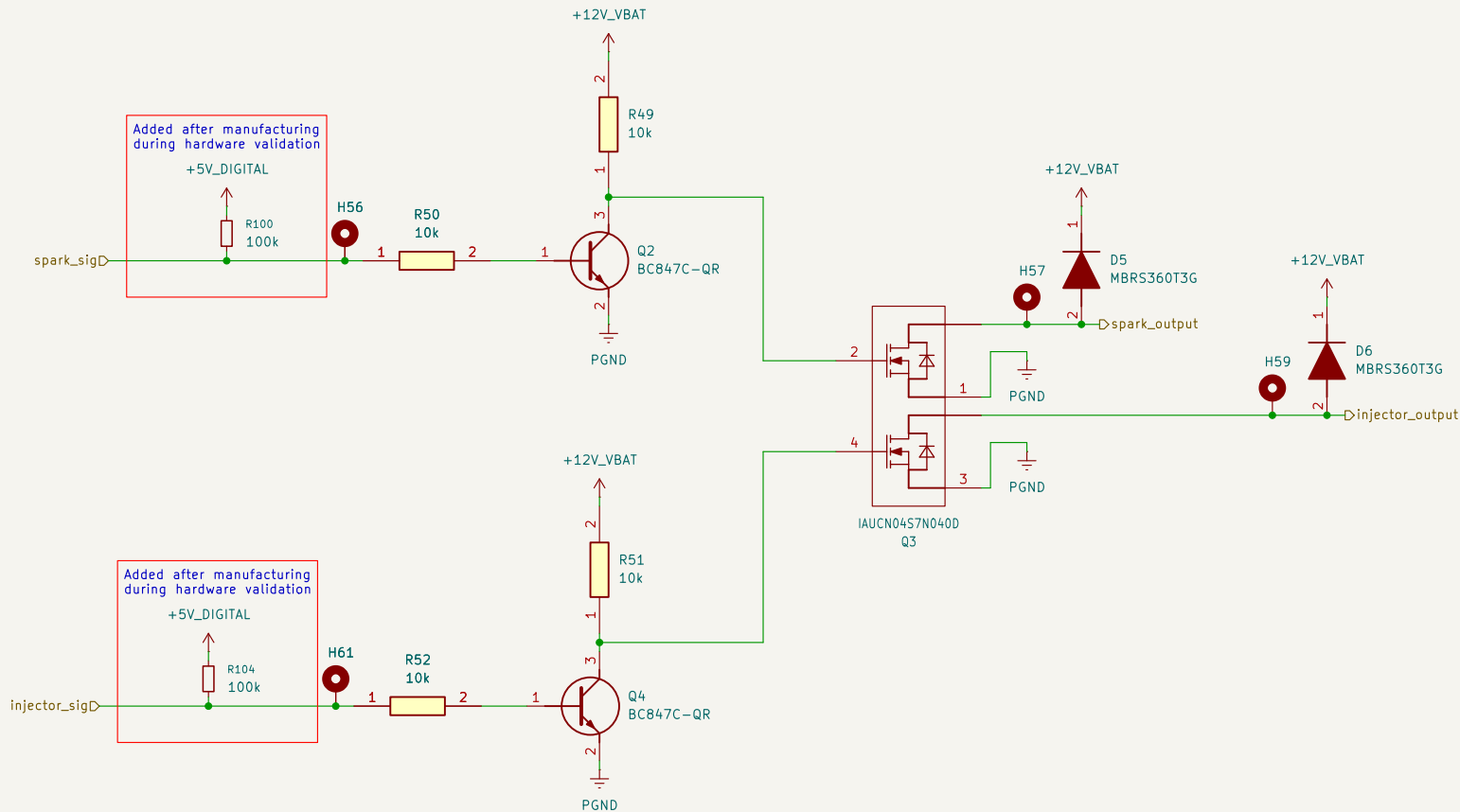
Rev: TimoUnu_A

KiCad E.D.A. 10.0.0

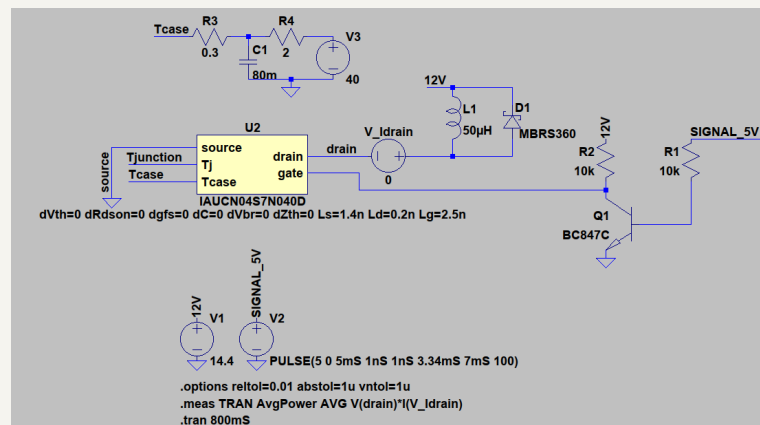
Id: 28/40

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Injection and Spark circuit for 1 cylinder



LTSice Simulation Schematic



Drawn by: Timotei-Iosif Cicu
Contact: timoteicicu1122@gmail.com
Checked by:

UPB Drive Formula Student Team

Sheet: /injection_and_spark_4_cylinder/injector_and_spark_1_cylinder1/
File: injection_and_spark_1_cylinder.kicad_sch

Title: TimoECU TimoUNU

Size: A4 Date: 2025-12-29

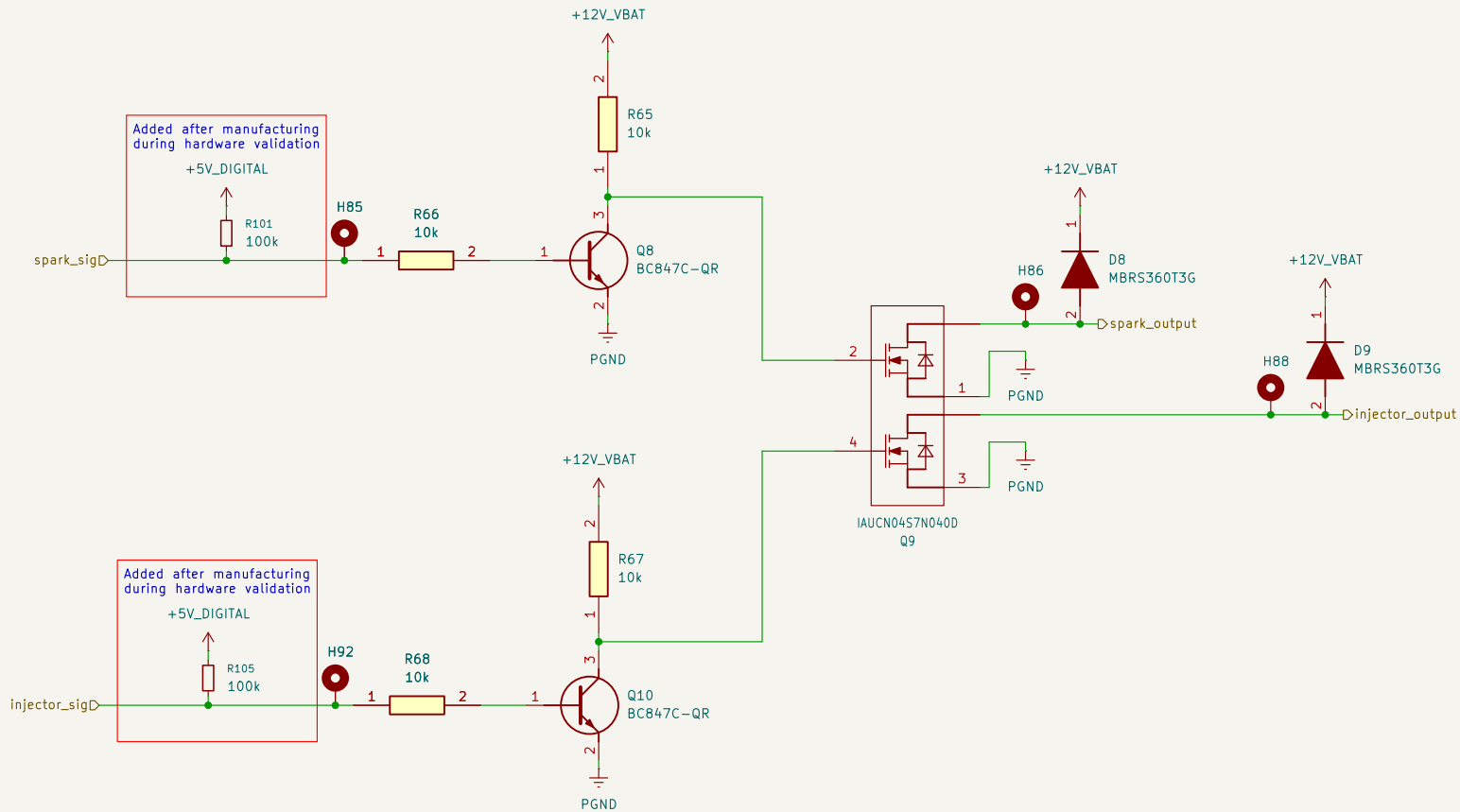
KiCad E.D.A. 10.0.0

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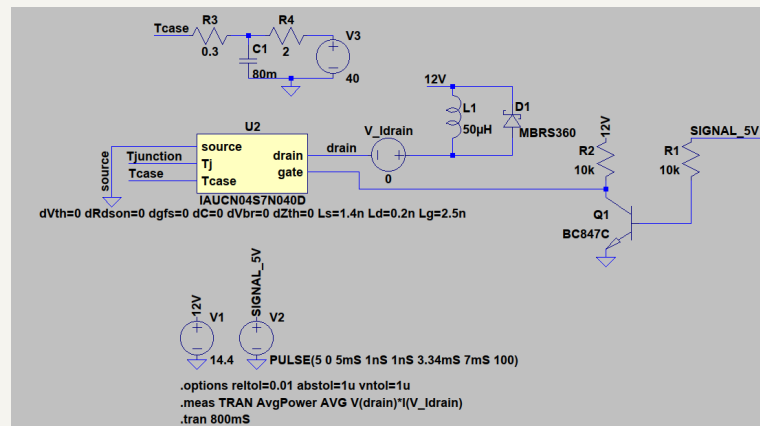
Rev: TimoUnu_A

Id: 27/40

Injection and Spark circuit for 1 cylinder



LTSice Simulation Schematic



Drawn by: Timotei-Iosif Cicu
Contact: timoteicicu1122@gmail.com
Checked by:

UPB Drive Formula Student Team

Sheet: /injection_and_spark_4_cylinder/injector_and_spark_1_cylinder2/
File: injection_and_spark_1_cylinder.kicad_sch

Title: TimoECU TimoUNU

Size: A4 Date: 2025-12-29

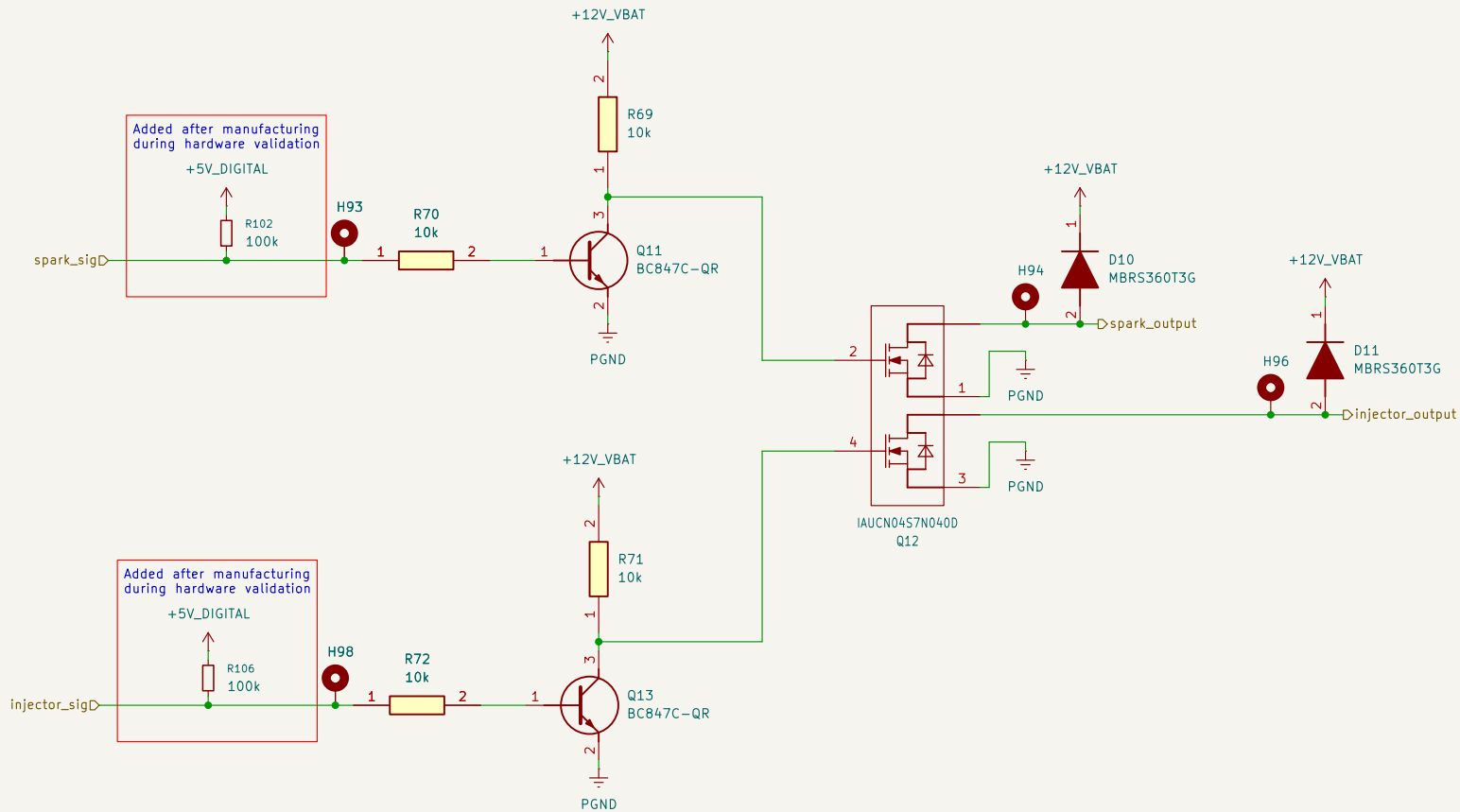
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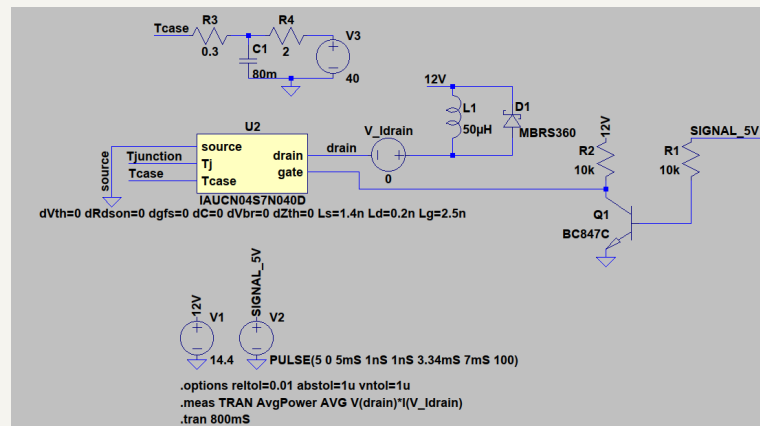
Rev: TimoUnu_A

Id: 36/40

Injection and Spark circuit for 1 cylinder



LTSice Simulation Schematic



Drawn by: Timotei-Iosif Cicu
Contact: timoteicicu1122@gmail.com
Checked by:

UPB Drive Formula Student Team

Sheet: /injection_and_spark_4_cylinder/injector_and_spark_1_cylinder3/
File: injection_and_spark_1_cylinder.kicad_sch

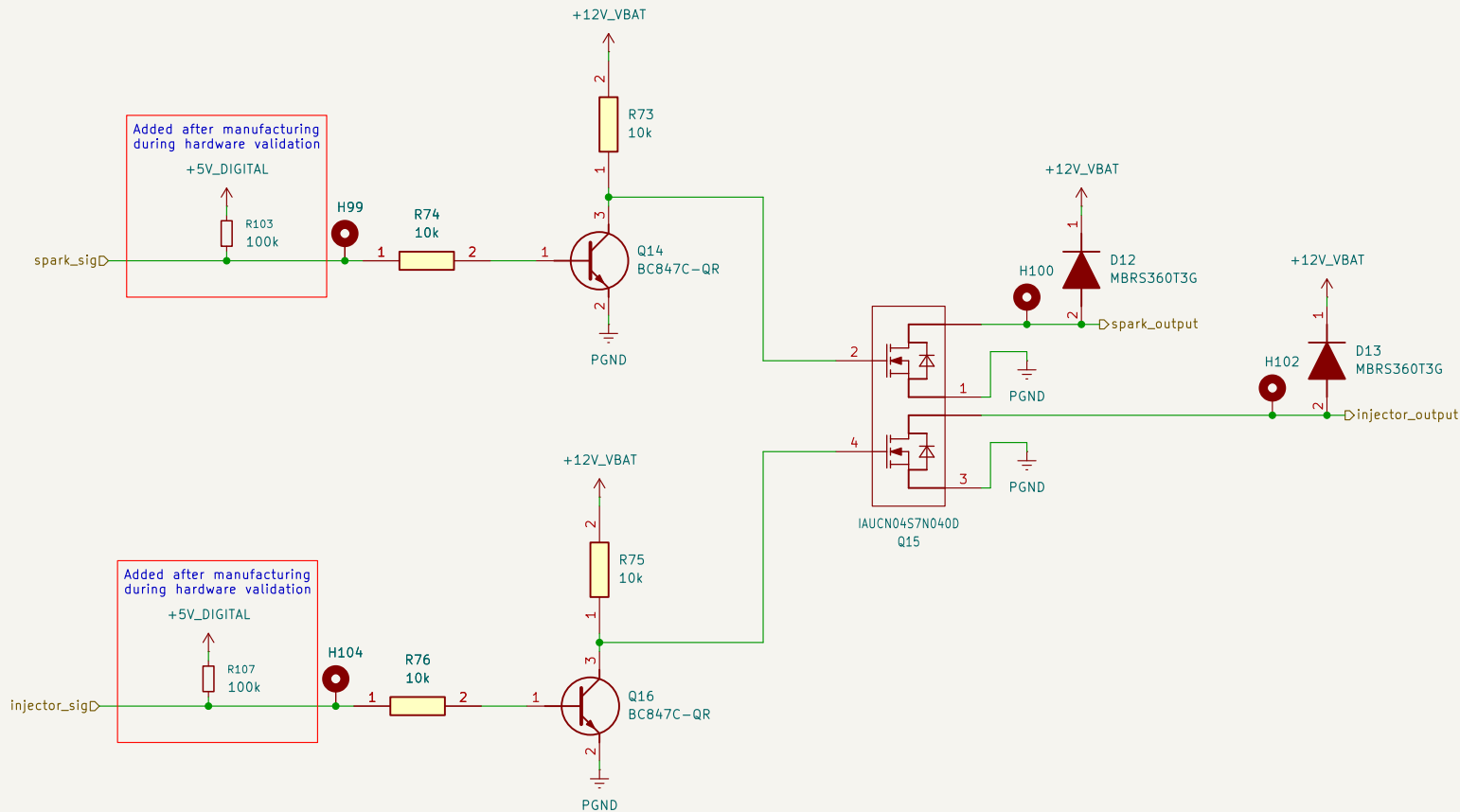
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Size: A4 Date: 2025-12-29
KiCad E.D.A. 10.0.0

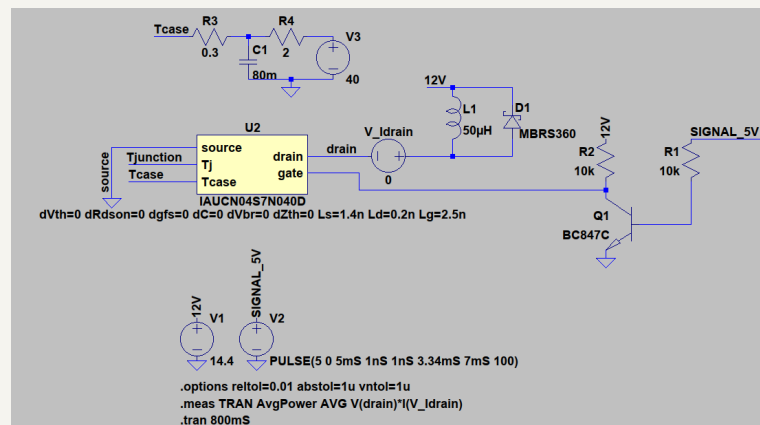
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Id: 37/40

Injection and Spark circuit for 1 cylinder



LTSice Simulation Schematic



Drawn by: Timotei-Iosif Cicu
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Sheet: /injection_and_spark_4_cylinder/injector_and_spark_1_cylinder4/
File: injection_and_spark_1_cylinder.kicad_sch

Title: TimoECU TimoUNU

Size: A4 Date: 2025-12-29

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Rev: TimoUnu_A

Id: 38/40

digital_output_sig_0

digital_output_sig

digital_output_0

File: digital_output.kicad_sch

digital_output_sig_1

digital_output_sig

digital_output_1

File: digital_output.kicad_sch

digital_output_sig_2

digital_output_sig

digital_output_2

File: digital_output.kicad_sch



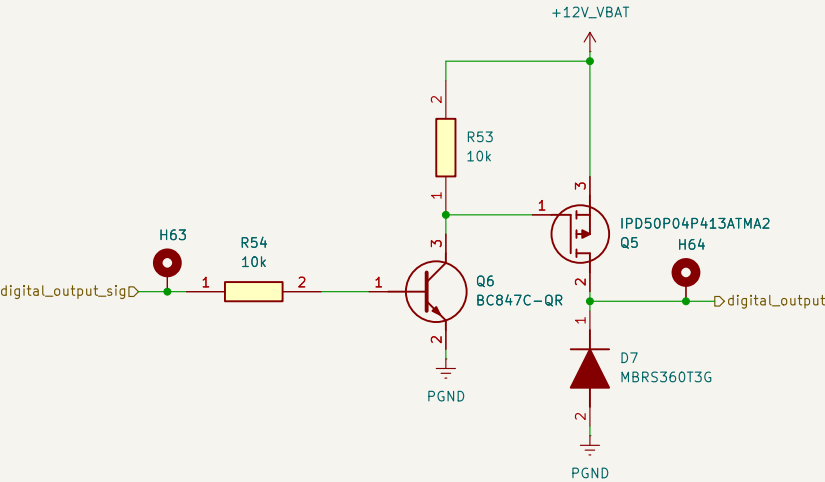
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Title: *TimoECU TimoUNU*

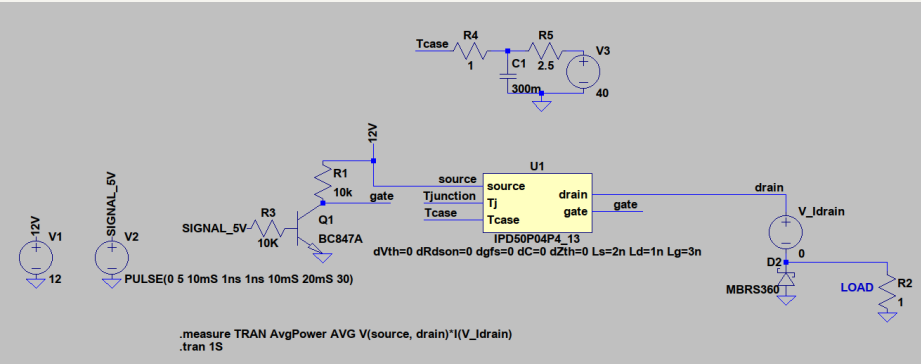
Rev: TimoUnu_A
Id: 29/40

Single Digital Output

KiCad Schematic



LTSpice Schematic



Drawn by: Timotei-Iosif Cicu
Contact: timoteicicu1122@gmail.com
Checked by:

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Sheet: /digital_outputs/digital_output0/
File: digital_output.kicad_sch

Title: TimoECU TimoUNU

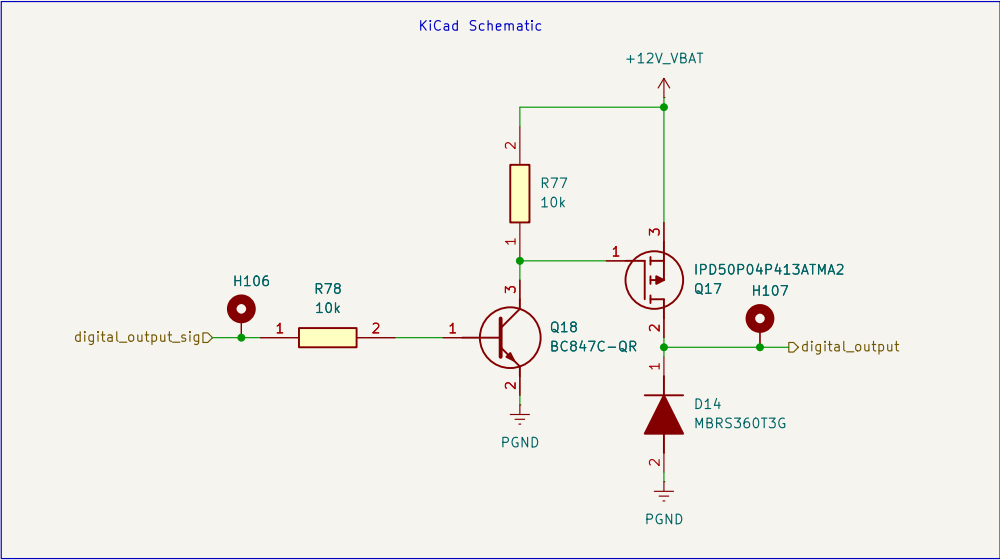
Size: A4 Date: 2025-12-29
KiCad E.D.A. 10.0.0

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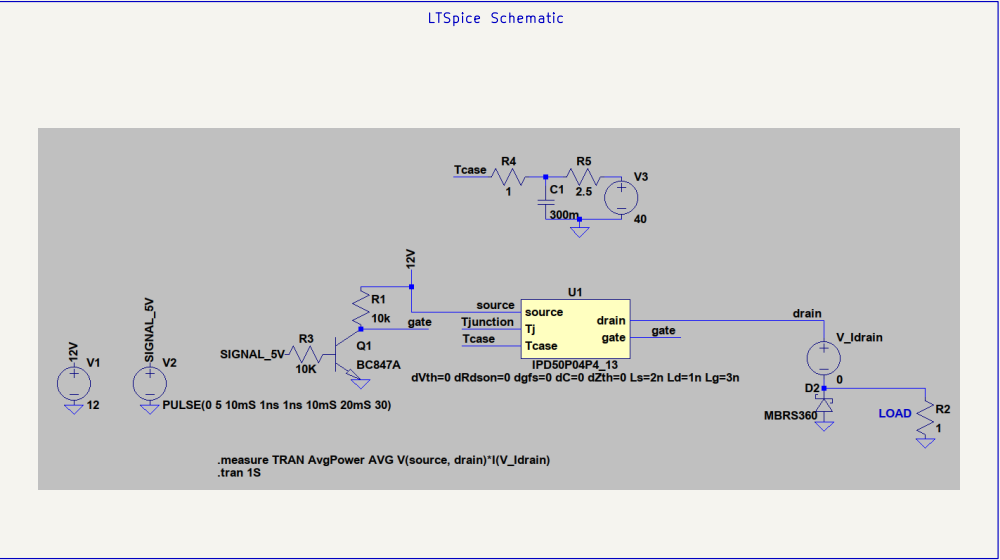
Rev: TimoUnu_A
Id: 30/40

Single Digital Output

KiCad Schematic



LTSpice Schematic



Drawn by: Timotei-Iosif Cicu
Contact: timoteicicu1122@gmail.com
Checked by:

UPB Drive Formula Student Team
Sheet: /digital_outputs/digital_output1/
File: digital_output.kicad_sch

Title: **TimoECU TimoUNU**

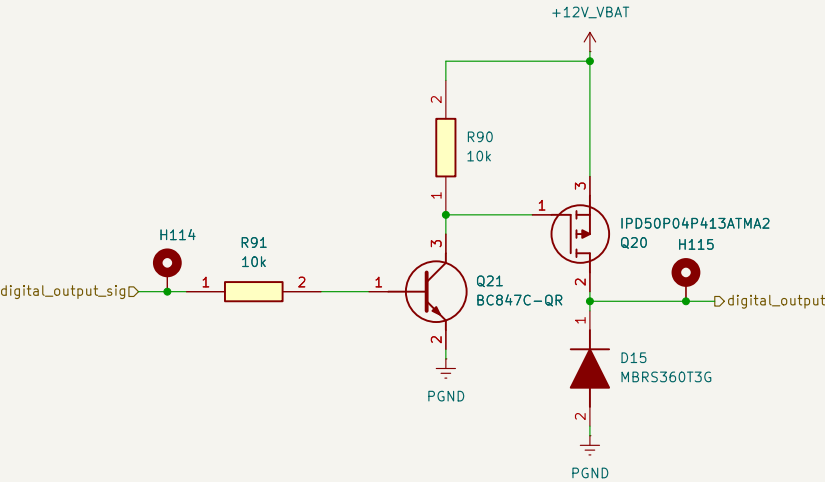
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KiCad E.D.A. 10.0.0

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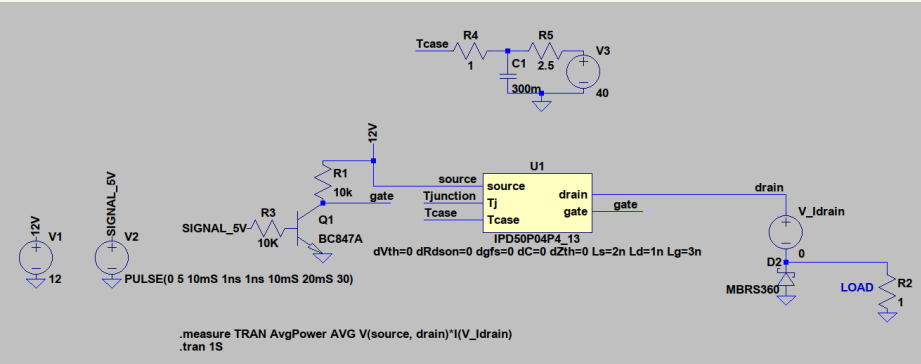
Rev: **TimoUnu_A**
Id: 39/40

Single Digital Output

KiCad Schematic



LTSpice Schematic



Drawn by: Timotei-Iosif Cicu
Contact: timoteicicu1122@gmail.com
Checked by:

UPB Drive Formula Student Team

Sheet: /digital_outputs/digital_output2/
File: digital_output.kicad_sch

Title: TimoECU TimoUNU

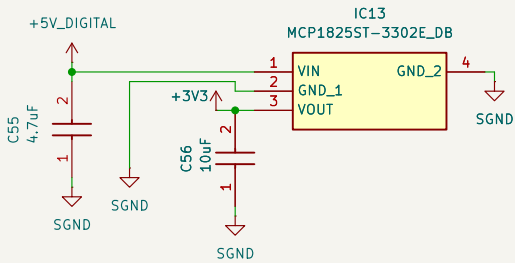
Size: A4 Date: 2025-12-29
KiCad E.D.A. 10.0.0

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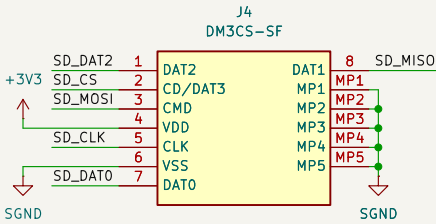
Rev: TimoUnu_A
Id: 40/40

SD Card Circuit

LDO for SD card



Card Holder

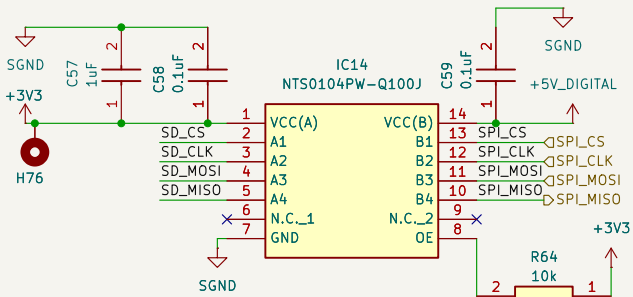


Note: Bypass capacitors already placed near level shifter

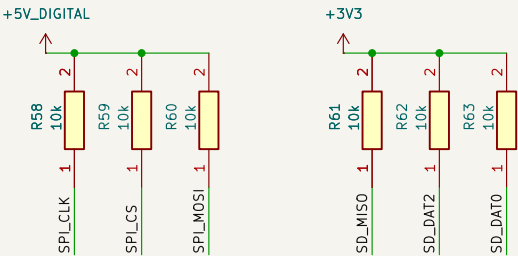
Reference schematic from commercial product

Note: Reference missing in released schematic due to it being from a datasheet or another person's project

Level Shifter 5V-->3.3V



Data lines pull-ups



Drawn by: Timotei-Iosif Cicu
Contact: timoteicicu1122@gmail.com
Checked by:
UPB Drive Formula Student Team
Sheet: /sd/
File: sd.kicad_sch

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FORMULA STUDENT TEAM

Title: **TimoECU TimoUNU**

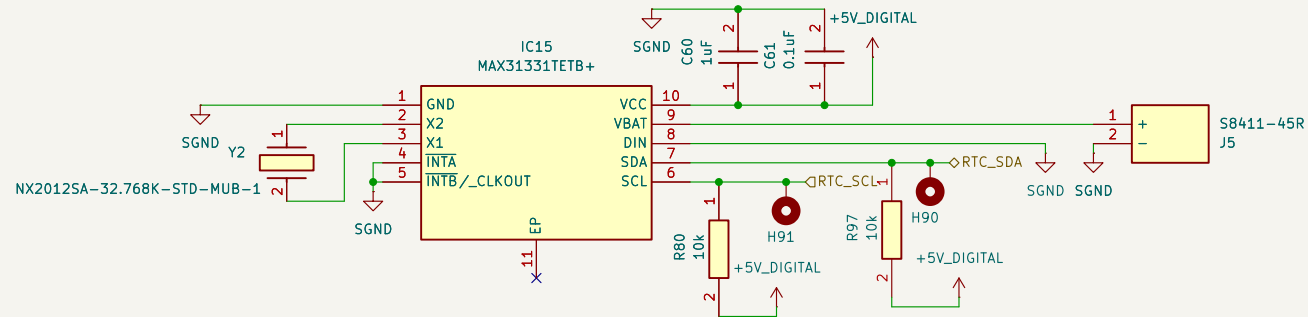
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Rev: **TimoUnu_A**

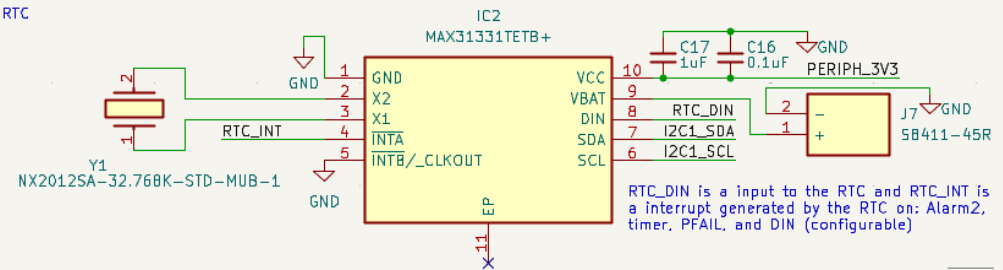
Id: 35/40

External RTC (S32K344 doesn't have internal RTC)



Reference from DR05's DAT

RTC



Drawn by: Timotei-Iosif Cicu
Contact: timoteicicu1122@gmail.com
Checked by:
UPB Drive Formula Student Team
Sheet: /rtc/
File: rtc.kicad_sch

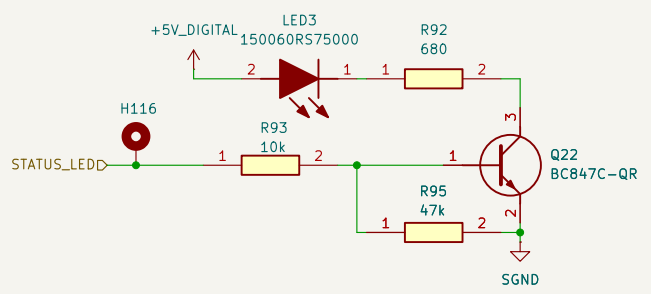
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Title: TimoECU TimoUNU

Size: A4 Date: 2025-12-29
KiCad E.D.A. 10.0.0

Rev: TimoUnu_A
Id: 36/40

Status LED



Drawn by: Timotei-Iosif Cicu
Contact: timoteicicu1122@gmail.com



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Sheet: /status_led/

File: status_led.kicad_sch

Title: TimoECU TimoUNU

Size: A4

Date:

Rev: TimoUnu_A

KiCad E.D.A. 10.0.0

Id: 41/40